

AIGR Gene Hypothesis Deep Research — Final Report

Target: Human ATG2A (UniProt Q2TAZ0) — GO:0034727 "piecemeal microautophagy of the nucleus"

Focus type: function_assignment · **Hypothesis slug:** function-hypothesis-go-0034727 **Source:** genes/human/ATG2A/ATG2A-ai-review.yaml (free-text) **Seed:** Human ATG2A participates in piecemeal microautophagy of the nucleus (GO:0034727), inherited via PANTHER node PTN000324023 from yeast ATG2.

Summary

Verdict: Over-annotated / refuted as a direct, applicable function assignment. The annotation of human ATG2A with GO:0034727 "piecemeal microautophagy of the nucleus" (PMN) is an IBA-only phylogenetic carry-over from budding yeast and should be treated as non-core and a candidate for removal in the ATG2A curated review. The human ATG2A GO:0034727 record carries evidence code IBA, assignedBy=GO_Central, reference=GO_REF:0000033 (PAINT phylogenetic annotation), and withFrom=PANTHER:PTN000324023 plus the yeast ATG2 gene (SGD:S000005186). That is precisely the phylogenetic propagation the seed hypothesis flagged. There is **no experimental (IDA/IMP/IGI/EXP) support** for PMN in human ATG2A, or in any metazoan.

The seed asked a sharp question: does ATG2A's lipid-transfer/membrane-tethering machinery participate in this *specific* microautophagic route in mammals, and is any absence a *demonstrated lineage restriction* or merely a gap in target experiments? The evidence points firmly to a **demonstrated lineage restriction**. PMN is a fungal-specific process defined at the **nucleus–vacuole (NV) junction**, which is built by the direct Nvj1–Vac8 interaction and requires a **vacuole**. QuickGO shows that **all 44 experimental-evidence PMN annotations are in *Saccharomyces cerevisiae* (100%)**. PANTHER v19 (positive-control validated by yeast ATG2 → human ATG2A/ATG2B) shows that **NVJ1 and VAC8 have no human ortholog**, while ATG2 is

conserved. Because the scaffolds that make PMN a distinct microautophagic route are absent from the human lineage, the term cannot biologically apply to human ATG2A — regardless of the fact that ATG2's own lipid-supply subunit is conserved.

Human ATG2A's genuine, experimentally supported function is **macroautophagic lipid transfer and membrane tethering** at ER–phagophore contact sites. Every direct-evidence GO annotation on Q2TAZ0 describes macroautophagy. The only mammalian nuclear-degradation events that are actually documented (lamin B1/LMNB1, SIRT1 turnover) are **LC3-dependent macroautophagic nucleophagy** — a *different* process and GO term from microautophagic PMN, and one whose dependence on ATG2A is itself untested. The recommended curation lead is therefore: **mark GO:0034727 non-core / remove**, retain the macroautophagy lipid-transfer core, and do **not** substitute a nucleophagy term without independent experimental support.

Key Findings

Finding F001 — ATG2A's direct/experimental function is macroautophagic lipid transfer; PMN is IBA-only

UniProt Q2TAZ0 (ATG2A, *Homo sapiens*) carries 17 GO cross-references. Every **experimental / direct-evidence** annotation describes **macroautophagy**, not microautophagy of the nucleus:

- GO:0120013 **lipid transfer activity** (IDA)
- GO:0000045 **autophagosome assembly** (IMP)
- GO:0034045 **phagophore assembly site membrane** (EXP)
- GO:0005789 **endoplasmic reticulum membrane** (EXP)
- GO:0044232 **organelle membrane contact site** (IDA)
- GO:2000786 **positive regulation of autophagosome assembly** (IDA)

By contrast, **GO:0034727 (PMN)** is annotated **only** as IBA:GO_Central — phylogenetic inference. It sits in a cluster of selective-autophagy *process* terms (pexophagy, reticulophagy, glycophyagy, mitophagy) that are all inherited by ancestry with no human experimental support. These are generic consequences of ATG2A being core autophagy machinery, not evidence of a demonstrated PMN role.

The core molecular activity is established directly. In [PMID: 31271352](#), *The autophagic membrane tether ATG2A transfers lipids between membranes*, the authors "demonstrate that human ATG2A is a lipid transfer protein. ATG2A can extract lipids from membrane vesicles and unload them to other vesicles," acting most efficiently when it tethers two membranes; recruitment is via the PI3P effectors WIPI4/WIPI1. [PMID: 33850023](#) frames ATG2 as the protein "proposed to transfer bulk lipid from the endoplasmic reticulum (ER) during autophagosome biogenesis" — i.e., ER-to-phagophore bulk lipid transfer during macroautophagy, with no requirement that ATG2 recognize nuclear cargo. This is reinforced by [PMID: 38622126](#), which shows ANKFY1 recruits ATG2A to PI3P-enriched endosomes to donate lipid to phagophores, again a macroautophagy function.

Interpretation. ATG2A's core, experimentally supported function is a lipid-transfer/membrane-tether activity feeding phagophore expansion in macroautophagy. PMN is not part of this experimentally supported core in humans.

Finding F002 — GO:0034727 has experimental support exclusively in *S. cerevisiae* via fungal-specific NV-junction machinery absent in mammals

QuickGO reports **59,856 total annotations** for GO:0034727, but restricting to experimental evidence (ECO:0000269 EXP and descendants) leaves **44 annotations, all in *S. cerevisiae* (taxon 559292; 100%)**. The experimentally annotated PMN gene set is a fungal cast: yeast ATG2 plus **NVJ1, VAC8, OSH1–7**, the core ATG1–18 machinery, and vacuolar fusion genes (VAM3/6/7, VPS41, YPT7). All non-yeast PMN annotations — including human ATG2A — are **IEA/IBA only**.

Mechanistically, PMN is degradation of the nucleus by **microautophagy** at **NV junctions** formed by the direct Nvj1 (perinuclear ER)–Vac8 (vacuole) interaction. [PMID: 18701704](#) states that "Piecemeal microautophagy of the nucleus (PMN) occurs in *Saccharomyces cerevisiae* at nucleus-vacuole (NV) junctions and results in the pinching-off and release into the vacuole of nonessential portions of the nucleus." [PMID: 28533415](#) shows that "Disruption of the Nvj1p–Vac8p interaction results in the loss of tight NVJs, which impairs piecemeal microautophagy of the nucleus" — the route is strictly junction-dependent. [PMID: 20943953](#) further shows PMN requires the vacuolar V-ATPase electrochemical gradient and Osh1p, and forms a vacuolar diffusion barrier — all fungal features.

Mammals have **no vacuole** and **no Nvj1/Vac8 orthologs** (Finding F003), so the specific microautophagic route named by GO:0034727 cannot occur in human cells. The mammalian nuclear-degradation events that *are* documented are **macroautophagic (LC3-dependent)**:

[PMID: 33292048](#) identifies "the nuclear lamina protein LMNB1 (lamin B1) as a nuclear autophagy substrate in primary human cells," and describes SIRT1 as a nuclear autophagy substrate — a process mechanistically distinct from yeast microautophagic PMN.

Interpretation. GO:0034727 is a **yeast-specific term** at the experimental level. Its cross-species propagation onto human ATG2A conflates a fungal microautophagic route with the (mechanistically unrelated) mammalian macroautophagic nucleophagy.

Finding F003 — PMN junction-defining proteins Nvj1 and Vac8 have no human ortholog, while ATG2 is conserved (PANTHER v19)

The single human ATG2A (UniProtKB:Q2TAZ0) GO:0034727 annotation is **evidence=IBA**, **assignedBy=GO_Central**, **reference=GO_REF:0000033** (PAINT), **withFrom=PANTHER:PTN000324023** and SGD:S000005186 (yeast ATG2) — pure phylogenetic propagation from yeast ATG2, exactly as the seed hypothesis anticipated.

An orthology test was run against the PANTHER v19 ortholog API (yeast taxon 559292 → human 9606), validated with a positive control:

Query protein (yeast)	UniProt	Human ortholog returned?	Result
ATG2 (positive control)	P53855	Yes — ATG2A (Q2TAZ0, least-diverged), ATG2B (Q96BY7)	Conserved
NVJ1 (NV-junction ER protein)	P38881	None	No human ortholog
VAC8 (NV-junction vacuolar protein)	P39968	None	No human ortholog

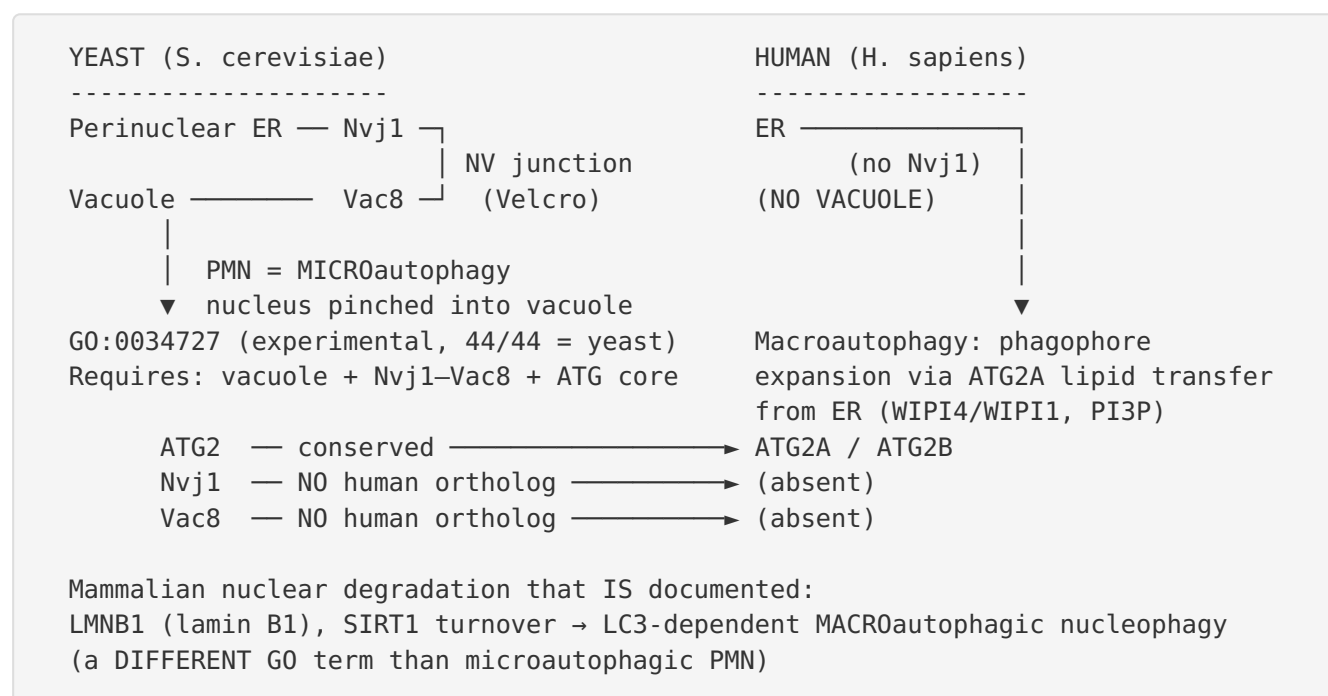
(An initial NVJ1 lookup mistakenly used P32602, which is actually SEC17/ α -SNAP; this was corrected to the verified NVJ1 accession P38881. All accessions were checked in UniProt.)

The two proteins that **define** the PMN NV junction have **no human ortholog**, while ATG2 itself is well conserved. Combined with F002, this resolves the seed's central question — "demonstrated lineage restriction or loss vs. absence of target experiments" — decisively toward **demonstrated lineage restriction**: the scaffold that makes PMN a distinct microautophagic route is absent from the human lineage.

Interpretation. ATG2 conservation is real, but it does not carry the PMN process with it, because PMN is defined by junction machinery (Nvj1/Vac8) and an organelle (the vacuole) that mammals lack. Propagating GO:0034727 to ATG2A on the strength of ATG2 orthology alone is a category error.

Mechanistic Model / Interpretation

The core issue is that **ATG2 conservation** $\neq$ **PMN conservation**. ATG2 is a shuttle/tether that supplies lipid to a growing membrane; the *identity of the degradative route* is set by other, non-conserved components.



Where ATG2A actually acts (human, experimentally supported):

Layer	Term	Evidence	Process
Molecular function	GO:0120013 lipid transfer activity	IDA	Macroautophagy
Cellular component	GO:0005789 ER membrane; GO:0034045 phagophore assembly site membrane; GO:0044232 organelle MCS	EXP/IDA	Macroautophagy
Biological process	GO:0000045 autophagosome assembly; GO:2000786 positive regulation of autophagosome assembly	IMP/IDA	Macroautophagy
Biological process	GO:0034727 PMN	IBA only	fungal microautophagy — not applicable

Even granting the seed's fair point that "lipid-supply participation does not require ATG2A to recognize nuclear cargo directly," the route still requires an NV junction and a vacuole to *be* PMN. Absent those, any ATG2A lipid-transfer contribution to a mammalian nuclear-degradation event would be captured by macroautophagy/nucleophagy terms, not by GO:0034727.

Evidence Base

Citation	Evidence type	Supports/ Refutes/ Qualifies/ Competing	Claim tested	Key finding	Context
PMID: 31271352	Direct in vitro assay	Refutes (redirects to macroautophagy)	ATG2A's core molecular function	Human ATG2A extracts/transfers lipids between vesicles; tether-enhanced; WIPI4/WIPI1-recruited	Recombinant human ATG2A + liposomes
PMID: 33850023	Model/review	Qualifies	ATG2 role is bulk ER→phagophore lipid transfer	Positions ATG2 in macroautophagy lipid supply with scramblases (TMEM41B/VMP1/ATG9)	Conceptual biochemistry
PMID: 38622126	Direct/interaction/mutant	Supports (macroautophagy)	ATG2A lipid-source flexibility	ANKFY1 recruits ATG2A to PI3P endosomes; ATG2A/B depletion impairs autophagosome growth	Human cells
PMID: 18701704	Mutant/morphological	Refutes (defines PMN as yeast-specific)	Where PMN occurs	PMN occurs in <i>S. cerevisiae</i> at NV junctions; requires core ATG genes	<i>S. cerevisiae</i>
PMID: 28533415	Structural + mutant	Refutes	PMN depends on Nvj1–Vac8 junction	Disrupting Nvj1p–Vac8p abolishes tight NVJs and impairs PMN	<i>S. cerevisiae</i>
PMID: 20943953	Mechanistic/imaging	Refutes	PMN vesicle formation requirements	NV junctions need Nvj1p/Vac8p, V-ATPase gradient, Osh1p; form vacuolar diffusion barrier	<i>S. cerevisiae</i>
PMID: 19182523	Morphological (EM)	Refutes	PMN genetics/morphology	PMN ("micronucleophagy") requires core ATG genes at NV junctions	<i>S. cerevisiae</i>

Citation	Evidence type	Supports/ Refutes/ Qualifies/ Competing	Claim tested	Key finding	Context
PMID: 15367582	Interaction/ localization	Refutes	PMN cofactors	Osh1p targeted to NV junctions via Nvj1p; Osh1–7 needed for PMN vesicles	<i>S. cerevisiae</i>
PMID: 15958487	Mutant	Refutes	PMN lipid requirements	VLCFA/Tsc13p at NV junctions shape PMN vesicle biogenesis	<i>S. cerevisiae</i>
PMID: 31512555	Structural/ biophysical	Refutes	Vac8 governs PMN vs Cvt	Vac8 quaternary states differentially regulate PMN and Cvt	<i>S. cerevisiae</i>
PMID: 33292048	Localization/ substrate	Competing (alternative route)	Mammalian nuclear autophagy mechanism	LMNB1 and SIRT1 are substrates of LC3-dependent (macro)nuclear autophagy	Primary human
PMID: 38308641	Mechanistic	Competing (alternative route)	Mammalian nucleophagy mechanism	SUMOylated Lamin B1 binds LC3 → lysosomal delivery (macroautophagic nucleophagy)	Rodent, human neuron
UniProt Q2TAZ0 / QuickGO annotation	Database	Refutes	Evidence basis of human PMN annotation	Human ATG2A GO:0034727 = IBA, GO_REF:0000033, withFrom PANTHER:PTN000324023 + yeast ATG2	Database
QuickGO experimental filter	Database/ computational	Refutes	Species distribution of PMN experiments	44/44 experimental GO:0034727 annotations are <i>S. cerevisiae</i>	Database

Citation	Evidence type	Supports/ Refutes/ Qualifies/ Competing	Claim tested	Key finding	Context
PANTHER v19 ortholog API	Computational/ evolutionary	Refutes	Conservation of PMN machinery	NVJ1, VAC8 → no human ortholog; ATG2 → ATG2A/ B (positive control)	Compu

GO Curation Implications (leads — require curator verification)

GO ID	Term	Aspect	On ATG2A now	Lead action
GO:0034727	piecemeal microautophagy of the nucleus	BP	IBA:GO_Central	REMOVE / treat as non-core — over-propagated, lineage-restricted (fungal) term
GO:0044804	autophagy of nucleus (nucleophagy)	BP	absent	Do not add without evidence (mammalian nucleophagy is macroautophagic; ATG2A role untested)
GO:0120013	lipid transfer activity	MF	IDA	RETAIN — core MF
GO:0000045	autophagosome assembly	BP	IMP	RETAIN — core BP
GO:2000786	positive regulation of autophagosome assembly	BP	IDA	RETAIN
GO:0034045	phagophore assembly site membrane	CC	EXP	RETAIN — core CC
GO:0005789	endoplasmic reticulum membrane	CC	EXP	RETAIN
GO:0044232	organelle membrane contact site	CC	IDA	RETAIN

Rationale. The experimentally supported core of ATG2A is a **lipid-transfer / membrane-tether MF** acting at **ER-phagophore contact sites** in **macroautophagy**. GO:0034727 is one of several IBA selective-autophagy process terms inherited by ancestry; for a fungal-specific *microautophagic* route absent from the human cell, the term is best treated as **not supported** /

non-core. We deliberately avoid recommending "protein binding" (GO:0005515) as a substitute; the informative core is the lipid-transfer MF and the macroautophagy CC/BP set, which are already present.

Mechanistic Scope

- Direct gene-product activity (human, experimentally supported):** intermembrane bulk **lipid transfer** and **membrane tethering**; ATG2A bridges ER (and PI3P endosomes, via ANKFY1) to the PI3P-marked phagophore, recruited by WIPI4/WIPI1, cooperating with scramblases (TMEM41B/VMP1/ATG9) to expand the phagophore. No nuclear-envelope targeting or nuclear-cargo recognition is documented
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P 31271352

P 33850023

P 38622126

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- PMN (yeast — the term's definitional scope):** microautophagic pinching of nuclear blebs directly into the vacuole at Nvj1–Vac8 junctions; core ATG genes (including yeast Atg2) are required for the terminal enclosure/release step — a lipid/membrane-machinery contribution, consistent with the seed's framing
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P 18701704

P 28533415

P 20943953

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- Downstream / different route (mammals):** macroautophagic nucleophagy of LMNB1 and SIRT1 (LC3-dependent). If ATG2A contributes here it does so as generic phagophore-lipid supply — its macroautophagy function — not as a dedicated nuclear program, and that contribution is currently untested
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P 33292048

P 38308641

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Separating the layers: the *immediate molecular function* (lipid transfer) is well supported and macroautophagic; PMN is a *cellular-route* claim that fails on machinery-conservation grounds; mammalian nucleophagy is a *distinct competing route* that would map to a different term.

Conflicts and Alternatives

1. **Cross-lineage database carry-over (primary confounder).** GO:0034727 experimental support is 100% *S. cerevisiae* (44/44); the human annotation is IBA (GO_REF:0000033/PAINT), withFrom PANTHER:PTN000324023 + SGD:S000005186 (yeast ATG2) — confirmed by QuickGO, exactly matching the seed. This is legitimate automated inference being tested for biological applicability.
2. **ATG2 orthology is real but insufficient.** ATG2 conservation (positive control: yeast ATG2 → human ATG2A/ATG2B) might *seem* to justify inheriting all yeast ATG2 processes. It does not, because PMN's identity depends on Nvj1/Vac8/vacuole, which are not conserved.
3. **Process-type mismatch.** The mammalian counterpart of nuclear autophagy is **macroautophagic**, not **microautophagic**
(
P 33292048

P 38308641
);
even a bona fide mammalian ATG2A nuclear role would map to a different GO term.
4. **Not paralog confusion.** ATG2A vs ATG2B redundancy concerns macroautophagy; both would inherit the same IBA PMN term. The issue is term specificity/lineage applicability, not paralog mis-assignment.
5. **Method single-sourcing.** The ortholog conclusion rests on PANTHER v19 (positive-control validated). Orthology inference is method-dependent (see Limitations).

Limitations and Knowledge Gaps

1. **Orthology inference is single-method.** NVJ1/VAC8 "no human ortholog" comes from PANTHER v19 with a validated positive control, but was not cross-checked here against OrthoDB, eggNOG, or InParanoid, nor by structure-based search. *Why it matters:* the central argument rests on machinery non-conservation. *Resolution:* confirm absence in ≥ 2

additional orthology resources and by Foldseek/structural search against the human proteome, since Nvj1/Vac8 are fast-evolving and sequence searches can miss remote homologs (Vac8 is an armadillo-repeat protein; humans have many ARM-repeat proteins, but functional NV-junction orthology is what matters).

2. **Does human ATG2A participate in macroautophagic nucleophagy?** Checked: no primary evidence found. A positive result would justify a *nucleophagy* (GO:0044804) lead — but not PMN. *Resolution*: ATG2A/B-KO + LMNB1/SIRT1 turnover assays.
3. **Is any microautophagy-of-nucleus route present in mammals at all?** Checked: literature shows only macroautophagic nuclear degradation. This bounds whether GO:0034727 could ever apply. *Resolution*: EM/live imaging of direct lysosomal engulfment of nuclear material.
4. **Absence of experiments ≠ proof of absence of function.** No study has tested whether ATG2A contributes lipid to any mammalian nuclear-membrane remodeling; the claim is specifically that the *PMN route* is inapplicable, not that ATG2A can never touch nuclear membranes.
5. **Database counts are time-stamped.** The 44/44 experimental-in-yeast figure reflects QuickGO at query time and could shift with future curation.

Discriminating Tests

- **ATG2A/ATG2B double-KO human cells + starvation/senescence:** quantify LMNB1/SIRT1 nucleophagic flux (autophagosome/lysosome) to test any *macronucleophagy* role (would support GO:0044804, not GO:0034727).
- **Multi-resource orthology + structural search:** confirm absence of Nvj1/Vac8 orthologs and vacuole-equivalent NVJ machinery in *Homo sapiens* via OrthoDB/eggNOG/InParanoid and Foldseek to formalize the lineage restriction.
- **Localization:** high-resolution imaging of endogenous ATG2A relative to the nuclear envelope during stress — absence of NE enrichment further argues against a dedicated nuclear route.
- **Literature sweep for metazoan PMN:** systematic keyword search for "piecemeal microautophagy of the nucleus" in any non-fungal organism (expected: none).
- **GO_Central PAINT review:** evaluate whether GO:0034727 IBA propagation should be taxon-constrained to Fungi at PANTHER node PTN000324023.

Proposed Follow-up Actions / Curation Leads (verify before applying)

- **Candidate action:** mark GO:0034727 on ATG2A as **NOT accepted / non-core (remove)**; label its basis as IBA over-propagation of a fungal-specific process, with an explicit note that NV-junction machinery (Nvj1/Vac8) and the vacuole are absent from the human lineage.
 - **Do not substitute** a broader nucleophagy/autophagy term for ATG2A on the basis of this annotation; any nucleophagy annotation would require independent experimental support.
 - **Retain** the macroautophagy lipid-transfer core (GO:0120013 IDA; GO:0000045 IMP; GO:2000786 IDA; GO:0034045/GO:0005789/GO:0044232 CC).
 - **Candidate references with exact snippets to cite/verify:**
 - [PMID: 31271352](#) — "human ATG2A is a lipid transfer protein. ATG2A can extract lipids from membrane vesicles and unload them to other vesicles" (core MF; retain).
 - [PMID: 28533415](#) — "Disruption of the Nvj1p-Vac8p interaction results in the loss of tight NVJs, which impairs piecemeal microautophagy of the nucleus" (PMN machinery is fungal).
 - [PMID: 18701704](#) — PMN "occurs in *Saccharomyces cerevisiae* at nucleus-vacuole (NV) junctions" (lineage restriction).
 - [PMID: 33292048](#) — "the nuclear lamina protein LMNB1 (lamin B1) as a nuclear autophagy substrate in primary human cells" (competing, macroautophagic route; different term).
 - Provenance: QuickGO — experimental PMN annotations = 44/44 *S. cerevisiae*; PANTHER v19 — NVJ1/VAC8 have no human ortholog.
 - **Suggested curator questions:** (a) Should PMN IBA propagation be taxon-restricted to Fungi at PANTHER node PTN000324023? (b) Is a generic *nucleophagy* (GO:0044804) lead warranted for ATG2A, or should nuclear-autophagy terms be withheld pending mammalian evidence?
 - **Suggested experiment:** ATG2A/B-KO nucleophagy flux assay (above).
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Bottom Line

Human ATG2A's demonstrated function is **macroautophagic lipid transfer/membrane tethering** at ER-phagophore contact sites. **GO:0034727 (piecemeal microautophagy of the nucleus) is a yeast-specific microautophagic process propagated to ATG2A by phylogeny**

only (IBA/PAINT, PANTHER:PTN000324023 + yeast ATG2); the defining machinery — a vacuole and the Nvj1–Vac8 junction — is absent in mammals (NVJ1/VAC8 have no human ortholog), a demonstrated lineage restriction rather than an untested cargo. Recommend treating GO:0034727 as over-annotated / non-core for human ATG2A (candidate for removal), while noting that a generic macroautophagic-nucleophagy role remains untested and must not be annotated without direct evidence.

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